

PROCESS DESIGN PRINCIPLES I — SHORT NOTES

CHE F314 û I-Semester 2026-27 û formulas as taught in the uploaded decks (Slides/)

Companion: `PDP-Problems-and-Solutions.pdf` (P1–P7, every question in the decks solved) and `pdp-answers-check.py`.

1 û Introduction and the hierarchy of decisions (L-1.pdf)

- Chemical process design: finding a sustainable process that converts raw materials to the desired products; a combination of science and art; synthesis (building the flowsheet) vs analysis (understanding a given one).
- Design problems are under-defined and open-ended (10^4 – 10^9 alternatives for one product): use heuristics (rules of thumb) first, shortcut methods where heuristics run out.
- Objective: lowest cost, safe, environmentally acceptable, operable.
- Hierarchy of decisions:** Level 1 batch vs continuous; Level 2 input–output structure; Level 3 recycle structure; Level 4 separation system (4a vapour recovery system VRS, 4b liquid separation system LSS); Level 5 energy-integration analysis (EIA).
- Batch is preferred for small capacities, campaign operation, fouling, or proven batch markets; continuous otherwise.

2 û Hierarchical approach, HDA case study (L-2.pdf)

- Synthesis steps:** (1) eliminate differences in molecular types (chemical reaction); (2) distribute chemicals by matching sources and sinks; (3) eliminate differences in composition (separation); (4) eliminate differences in T , P and phase; (5) integrate tasks into unit operations.
- HDA process (toluene + H_2 → benzene + CH_4): reactor at 1150–1300 °F (below: too slow; above: hydrocracking), 500 psia (≈ 34 atm), excess H_2 to avoid coking; purge needed because CH_4 builds up; toluene recycled.
- Boiling points guide the separation: H_2 –423, CH_4 –259, benzene 176, toluene 232, diphenyl 491 °F ⇒ flash/light-gas split first, then distillation.
- Ordering rules: the VRS is designed *before* the LSS (its losses fix the purge composition); EIA is done *last*, after the distillation train is fixed (HENS needs every stream's flow, composition and temperature).
- Input–output structure dominates economics: raw materials are normally 33–85 % of total product cost; overall material balances are evaluated first, and alternatives whose products are worth less than their feeds are discarded immediately.

3 û Pathway economics and the VCM case study (L-4.pdf)

- Gross profit = value of products – cost of reactants, per lb of vinyl chloride (VC); path comparison uses lb/lb VC from stoichiometry × (cents/lb) prices (ethylene 18, acetylene 50, Cl_2 11, VC 22, HCl 18, water 0).
- Path-3 flowsheet (balanced for 100 000 lb/h VC): Cl_2 113 400 lb/h, C_2H_4 44 900 lb/h, DCE to pyrolysis 158 300 lb/h (60% per-pass conversion ⇒ 105 500 lb/h DCE recycle); chlorination 100% conversion.
- Pressure decisions: chlorination 1.5 atm (no air ingress); pyrolysis 26 atm; first column 12 atm (HCl boils at –26.2 °C, bottoms bubble point 93 °C, away from the critical region); second column 4.8 atm (distillate 33 °C condenses on cooling water, bottoms 146 °C on medium-pressure steam); feed ≥ 35 °C.
- Pump liquids, not vapours (molar volume); cool the pyrolysis effluent by partial condensation to set the feed condition.
- Flow diagrams:** BFD (functional blocks) → PFD (all major units, numbered stream diamonds, stream table, utilities, equipment summary) → P&ID (every instrument and control loop; the construction document).

4 û Economic decision making: the acetone case study (EDM.pdf)

- Economic potential EP = product value – raw-material cost; here 10.3 mol/h acetone in 687 mol/h air ⇒ EP = Rs 5.26 Cr/yr ($\approx \$1.315 \times 10^6$ /yr at 8150 h/yr).
- Recovery alternatives: condensation (high P /low T), absorption, adsorption, membrane, reaction. Heuristic: solute < 5 mol% in the gas ⇒ adsorption cheapest (here 10.3/697.3 = 1.47 %).
- Heuristics:** H1 – if a raw-material component can act as the absorber solvent (water), consider feeding the process through the gas absorber (justifies discarding part of the process water, \$25 600/yr vs EP). H2 – recover > 99 % of valuable material (first guess 99.5 %). H3 – isothermal dilute absorber: $L = 1.4 mG$. H4 – avoid adiabatic absorbers (their minimum L can be $\sim 10\times$ larger, $L \approx 14 mG$).
- Equilibrium slope $m = \gamma_s P_s^\circ / P$: acetone–water at 25 °C, 1 atm: $m = 6.7 \times 229/760 = 2.02$; $L = 1.4 mG = 1943$ mol/h.
- Kremser equation** (isothermal, dilute, straight lines): $N + 1 = \frac{\ln \left[\left(\frac{y_{in} - mx_{in}}{y_{out} - mx_{in}} \right) \left(1 - \frac{mG}{L} \right) + \frac{mG}{L} \right]}{\ln(L/mG)}$; with $L/mG = 1.4$, $x_{in} = 0$: 99 % recovery ⇒ $N \approx 10$ (exact 10.1), 99.9 % ⇒ 16; the back-of-envelope form $N \approx \log(y_{in}/y_{out}) / \log(L/mG)$ type simplification reproduces this. Keep $1 < L/mG < 2$ (1.4 is the trade-off; $L/mG < 1$ needs infinite trays, > 2 starves the distillation column feed).
- Optimum solute loss from the TAC model $TAC = C_s G y_{in} \left(\frac{y_{out}}{y_{in}} \right) 8150 + C_N [N + 2]$: $\frac{y_{out}}{y_{in}} \approx \sqrt{\frac{C_N \ln(L/mG)}{8150 C_s G y_{in}}}$ -type result;

numerically 0.004 (i.e. 99.6 %) and insensitive to $\pm 100\%$ cost changes.

- Design-variable effects: doubling P_T halves L (smaller still, same trays); hotter solvent raises $\gamma, P_s^\circ \Rightarrow m, L$; MIBK ($\gamma = 1$) shrinks L but its own cost kills it.
- **Approach-temperature heuristic:** 10°F from ambient to the boiling point of organics; the deck's own evaluation (waste stream $F = 51\,000\text{ lb/h}$, $c_p = 1.0$, $U_c = 30$, $U_s = 20\text{ Btu/h}\cdot\text{ft}^2\cdot^\circ\text{F}$, $C_A = \$11.38/\text{yr}\cdot\text{ft}^2$) gives an optimum near 21.5°F — the rule is conservative (see P1).
- **Back-of-envelope distillation:** $N_{min} \approx \frac{T_{b1} + T_{b2}}{3(T_{b2} - T_{b1})}$ is essentially Fenske's equation $N_{min} = \frac{\log \left[\frac{x_D}{1-x_D} \frac{1-x_B}{x_B} \right]}{\log \alpha}$ when $\lambda/R \approx 5080\text{ K}$ (see P2).

5 û Energy integration (EIA_Selected_Slides.pdf)

- Hot and cold streams are drawn as enthalpy–temperature composite curves; the closest vertical distance is ΔT_{min} (the deck works with the 10°F approach).
- The pinch (where the curves touch at ΔT_{min}) splits the problem: no heat crosses the pinch; above it only hot utility, below it only cold utility.
- Exchanger duty and area: $Q = U A \Delta T_{lm}$, so $A = \frac{Q}{U \Delta T_{lm}}$ with $\Delta T_{lm} = \frac{\Delta T_1 - \Delta T_2}{\ln(\Delta T_1/\Delta T_2)}$.
- The grid diagram records matches: exchangers as circles/lozenges between stream lines, heaters and coolers at the ends; matches that cross the pinch are the retro-fit targets.

Compiled from the uploaded decks; **PDP-Problems-and-Solutions.pdf** applies these formulas to every numerical the decks pose (P1–P7), and **pdp-answers-check.py** reproduces all numbers.